Practice Exam Numerical Recipes for Astrophysics A

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List your name and student number on ALL answer sheets

Total number of points: 40
Minimum score to pass: 22

This exam is divided into three parts: straightforward knowledge questions, questions that require understanding and take a bit more thought, and finally questions that have you apply your knowledge and understanding to a set of hypothetical or concrete test problems.

Pen, ruler, pencil, eraser and calculator are recommended. When asked for a definition or explanation, "in your own words" is implied.

Textbooks, notes, phones and graphic calculators are not allowed.

You are required to hand in this exam along with your solutions.

Useful equations

- Probability density of the normal distribution:

$$p(x|\mu,\sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Probability density of the Poisson distribution:

$$p(k|\lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$$

- Derivative of a power:

$$\frac{\partial(x^y)}{\partial y} = x^y \ln(x)$$

- Taylor expansion around $x = a$:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2 + \dots$$

Exam questions

1) Knowledge (1 point each)

(a) Describe the Metropolis-Hastings algorithm. In which method is it used?

(b) Define open, semi-open and closed methods of integrating functions.

(c) How many function points do you need for 1D polynomial interpolation of degree k?

(d) In the context of solving linear equations, what is pivoting?

(e) What is the smallest absolute error we can reasonably expect on recovering a minimum at $x = b$, $|x - b|$, in terms of the machine error ϵ_{m}?

2) Understanding (2 points each)

(a) `n` is an unsigned integer. What mathematical operation is equivalent to `n & 15`?

(b) Below are two different ways of calculating the same polynomial. Which is more efficient, and most importantly, why?

$$p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots + a_{\mathrm{n}} x^n$$
$$p(x) = a_0 + x(a_1 + x(a_2 + x(a_3 + \cdots + x(a_{\mathrm{n}-1} + a_{\mathrm{n}} x) \ldots)))$$

(c) Give an example of a robust estimator. Why does this give a more robust estimate than e.g. a least squares or χ^2 for certain data?

(d) Describe how Romberg integration is able to yield a much smaller error with a small number of evaluations than a simpler method (e.g. extended trapezoid or Simpson's rule) with a large number of evaluations.

(e) In a typical scenario, quicksort is about the quickest way to sort. Name two situations in which you might prefer e.g. mergesort over it anyway.

(f) How does the 64-bit XOR-shift method work? Argue why it's better or worse than the linear congruential generator.

(g) Say that we have a function that returns a uniform random integer between 0 and 2^{32} (exclusive). How can we use this to get a set of uniformly random floating point numbers between 5 and 12 (exclusive)?

(h) Any numerical solution to a system of linear equations will have some error due to e.g. the propagation (and possibly magnification) of the machine precision. How can we reduce the error in the solution after calculating it?

(i) The probability distribution relevant for χ^2 values is given by:

$$p(x, k) = \left[2^{k/2}\Gamma(k/2)\right]^{-1} x^{k/2-1} e^{-x/2}.$$

Which quantities are represented by x and k? How can we use this probability distribution to evaluate our fit?

(j) Explain what a natural cubic spline is. What are the constraints that fix its parameters?

3) Application (3 points each)

(a) Give a set of steps to efficiently and accurately numerically calculate the following integral:

$$\int_{-5}^{\infty} |z|^{-1/2} e^{-z/2} \, \mathrm{d}z.$$

Note: you don't actually have to calculate it.

(b) Given are four function values $f(x, y)$ in 2D, at (x_0, y_0), (x_1, y_0), (x_0, y_1) and (x_1, y_1). Show or derive how bilinear interpolation works and give a bilinearly interpolated value at (x, y), interior to the four points.

(c) Below is an array with 7 elements. Show how the quicksort algorithm would sort this array by giving each state of the array from unsorted to sorted, briefly describing what the algorithm is doing between states.

| 5 | 7 | 3 | 2 | 4 | 9 | 1 |

(d) Consider Figure 1, showing the contours of a 2D function as well as an initial simplex. We now call the downhill simplex method on it with a maximum number of iterations equal to 5. Show how the method proceeds by drawing in the figure and clearly indicate the minimum returned after 5 iterations. Mention what happens at each iteration.

(e) Given in Table 1 is a list of 20 random pairs of numbers (x, y), with $x, y \sim U(0, 1)$. Using these numbers, estimate the area of the shaded region in Figure 2. What algorithm are you using? **BONUS:** Give an uncertainty estimate on the area you calculated.

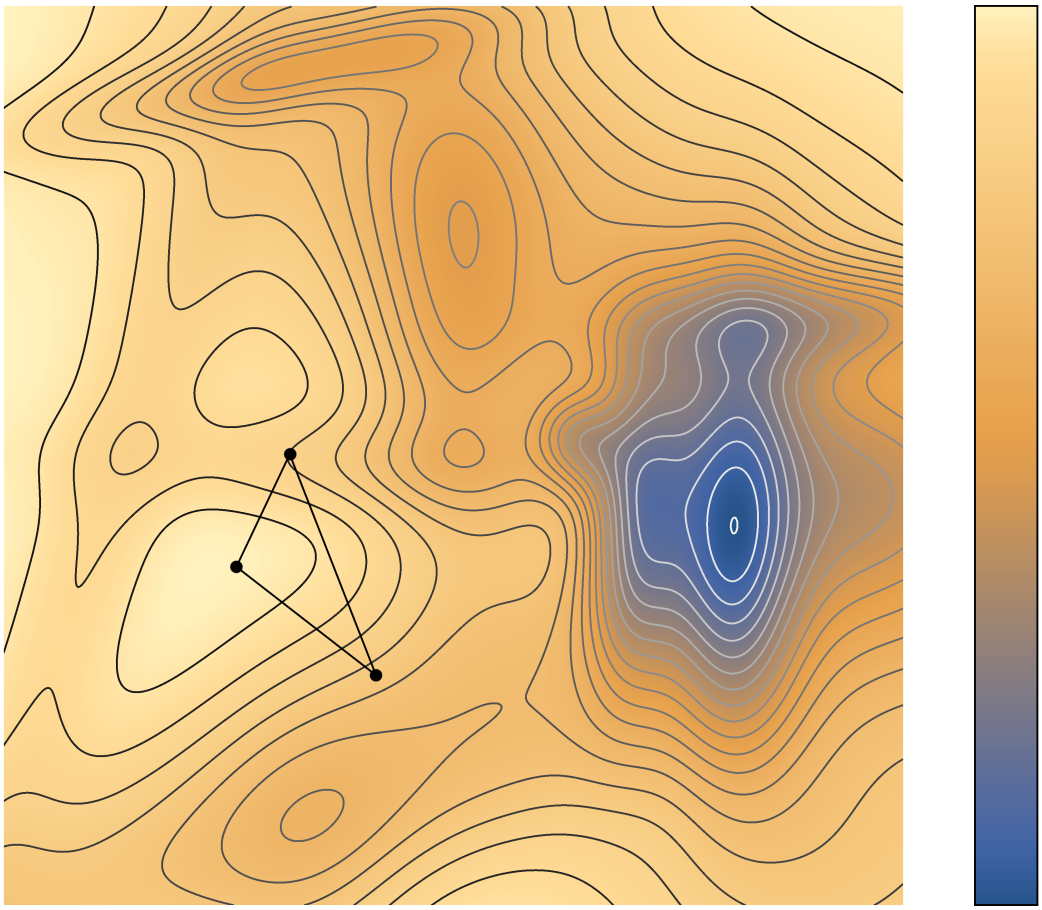

Figure 1: See question 3(d).

x	0.813	0.728	0.943	0.825	0.868	0.980	0.338	0.990	0.710	0.170
y	0.254	0.831	0.235	0.357	0.107	0.919	0.915	0.458	0.016	0.316
x	0.230	0.638	0.685	0.510	0.490	0.228	0.680	0.810	0.853	0.965
y	0.592	0.558	0.780	0.441	0.851	0.308	0.594	0.089	0.501	0.411

Table 1: See question 3(e).

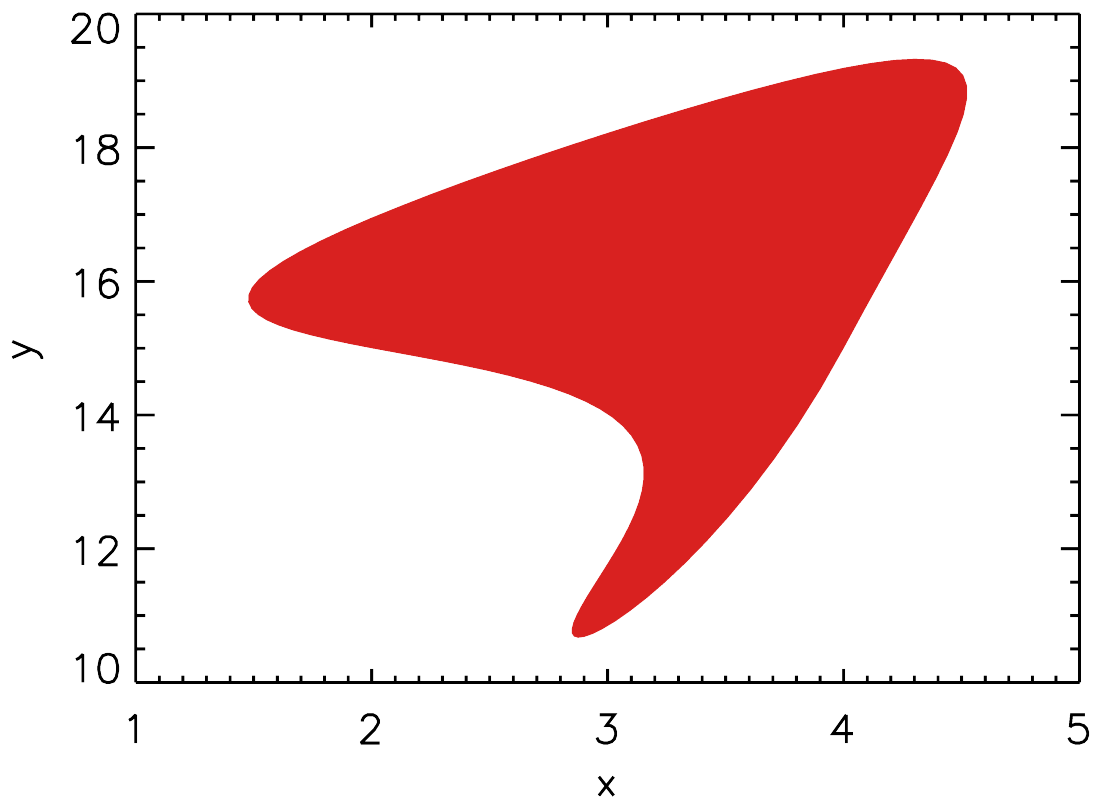

Figure 2: See question 3(e).